ROTATION INVARIANCE FROM LATTICE SYMMETRY AND REFLECTION POSITIVITY

Abstract. A Gaussian-dominated law of closed two-forms on $\mathbb{R}^d$, $d \geq 3$, is invariant under every Euclidean isometry if it is invariant under translations and signed coordinate permutations and reflection positive across a coordinate hyperplane and a diagonal hyperplane. We measure the failure of rotation invariance by the difference between a moment and its rotated moment. The two reflection directions give the spectral inequality $H \geq |P|$. Together with Gaussian domination, this yields analytic continuation and bounds the angular frequencies of the difference. Reflection positivity in the coordinate and diagonal frames then eliminates every nonzero frequency. Closedness ensures that a defect cannot be confined to coincident planar points.

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1. Introduction

Fix an integer $d \geq 3$. For $0 \leq j \leq d-1$, let x_j be the jth Cartesian coordinate on $\mathbb{R}^d$, and write

$$t = x_0, \qquad x = x_1, \qquad y = (x_2, \dots, x_{d-1}) \in \mathbb{R}^{d-2}. \tag{1}$$

Let $\mathcal{S}_2 = \mathcal{S}(\mathbb{R}^d; \Lambda^2\mathbb{R}^d)$ be the Fréchet space of real two-forms

$$\chi = \sum_{0 \leq j < k \leq d-1} \chi_{jk} \, \mathrm{d}x_j \wedge \mathrm{d}x_k$$

with Schwartz coefficients. We identify forms and their coefficient duals using the Euclidean metric. Thus the continuous dual $\mathcal{S}_2'$ consists of distributional two-forms, with pairing and test norm

$$F(\chi) = \sum_{j<k} \langle F_{jk}, \chi_{jk} \rangle, \qquad \|\chi\|_2^2 = \sum_{j<k} \int_{\mathbb{R}^d} |\chi_{jk}(q)|^2 \, \mathrm{d}q.$$

The pairing extends complex-linearly to complex tests.

Let μ be a Borel probability measure on $\mathcal{S}_2'$, equipped with its weak dual topology. Its characteristic function is

$$\Phi(\chi) = \int_{\mathcal{S}_2'} e^{iF(\chi)} \, \mathrm{d}\mu(F). \tag{2}$$

We write

$$S_N(\chi_1, \dots, \chi_N) = \int_{\mathcal{S}_2'} F(\chi_1) \cdots F(\chi_N) \, \mathrm{d}\mu(F), \qquad S_0 = 1, \tag{3}$$

for its moments whenever they exist.

We use T^* for the ordinary pullback. For an isometry $T(q) = Rq + a$,

$$(T^*\chi)_{jk}(q) = \sum_{\ell<m} (R_{\ell j}R_{mk} - R_{mj}R_{\ell k})\chi_{\ell m}(Tq). \tag{4}$$

This action preserves the L^2 norm. If Θ is reflection across an affine hyperplane, then μ is *reflection positive* across that hyperplane if

$$\sum_{\alpha,\beta=1}^{m} \overline{z_\alpha} z_\beta \, \Phi(\chi_\alpha - \Theta^*\chi_\beta) \geq 0 \tag{5}$$

for every $m \geq 1$, every $z_1, \dots, z_m \in \mathbb{C}$, and all real tests χ_α supported on the same side of the hyperplane.

Theorem A. *Suppose that μ satisfies the following four properties:*

(i) $dF = 0$ for μ-almost every $F \in \mathcal{S}_2'$.

(ii) There is a constant $C > 0$ such that, for every real $\chi \in \mathcal{S}_2$,

$$\int_{\mathcal{S}_2'} e^{F(\chi)} \, \mathrm{d}\mu(F) \leq e^{C\|\chi\|_2^2}. \tag{6}$$

(iii) $\Phi(T^*\chi) = \Phi(\chi)$ whenever $T(q) = Rq + a$, where $a \in \mathbb{R}^d$ and R is a signed permutation matrix.

(iv) μ is reflection positive across

$$H_\circ = \{t = 0\}, \qquad H_\Delta = \{t + x = 0\}. \tag{7}$$

Then μ is invariant under every Euclidean isometry:

$$\Phi(T^*\chi) = \Phi(\chi) \qquad (T \in \mathrm{Iso}(\mathbb{R}^d),\ \chi \in \mathcal{S}_2). \tag{8}$$

It is also reflection positive across every affine hyperplane. All joint moments exist, extend continuously to complex L^2 two-form tests, and satisfy

$$S_N(T^*\chi_1, \dots, T^*\chi_N) = S_N(\chi_1, \dots, \chi_N) \tag{9}$$

for every $N \geq 1$ and every $T \in \mathrm{Iso}(\mathbb{R}^d)$.

A signed permutation matrix has one nonzero entry, equal to 1 or -1, in each row and column. The assumed symmetries transport reflection positivity from $H_\circ, H_\Delta$ to all coordinate and diagonal hyperplanes, including their translates. We denote the two displayed reflections by $\Theta_\circ, \Theta_\Delta$.

The closedness assumption will enter through tests. If d^* is the formal L^2 adjoint of d, then

$$F(d^*\eta) = 0 \qquad (\eta \in C_c^\infty(\mathbb{R}^d; \Lambda^3\mathbb{R}^d)). \tag{10}$$

Our sign convention is $(d^*\eta)_{i_1\cdots i_{k-1}} = -\sum_j \partial_j \eta_{j i_1 \cdots i_{k-1}}$. Pullback by an isometry commutes with both d and d^*.

Gaussian domination gives a bound

$$|S_N(\chi_1, \dots, \chi_N)| \leq (K\sqrt{N})^N \prod_{j=1}^{N} \|\chi_j\|_2, \tag{11}$$

where K depends only on C. The proof is in Appendix A. In particular, the moments have distribution kernels. When some coordinates are left unsmeared below, we continue to use S_N for these kernels, specifying their arguments explicitly.

The analytic strategy belongs to the finite-covariance method of Bros, Epstein, and Glaser [BEG], Streater [S], and Osipov [O]. The spectral comparison and field estimates below follow the supplied manuscript [M, §18]; Sokal also proposed using diagonal spectral representations to restore covariance [So, §5]. We include the needed estimates. No model-specific covariance theorem is being applied to the law in Theorem A.

2. Proof sketch

The group $O(d)$ is generated by signed permutation matrices and the rotations

$$R(\theta) = \begin{pmatrix} r(\theta) & 0 \\ 0 & I_{d-2} \end{pmatrix}, \qquad r(\theta) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}. \tag{12}$$

The signed permutations are already symmetries. It therefore suffices to prove invariance under rotations of the tx-plane.

For an N-tuple of tests $\chi = (\chi_1, \dots, \chi_N)$, define

$$D[\chi](\theta) = S_N(R(\theta)^*\chi_1, \dots, R(\theta)^*\chi_N) - S_N(\chi_1, \dots, \chi_N). \tag{13}$$

Our task is to prove $D[\chi](\theta) = 0$ for every χ and θ.

2.1. **The Fourier coefficients of the defect.** For Schwartz tests, $D[\chi]$ is smooth: rotations depend smoothly on θ in the Schwartz topology, and S_N is continuous. Hence

$$D[\chi](\theta) = \sum_{\ell\in\mathbb{Z}} c_\ell[\chi] e^{i\ell\theta}, \qquad c_\ell[\chi] = \frac{1}{2\pi}\int_0^{2\pi} D[\chi](\theta) e^{-i\ell\theta}\,\mathrm{d}\theta. \tag{14}$$

Only $\ell \in 4\mathbb{Z}$ occur, since a quarter turn is an assumed symmetry. For real tests, $c_{-\ell}[\chi] = \overline{c_\ell[\chi]}$. It is enough to eliminate the nonzero frequencies: the remaining constant then vanishes because $D[\chi](0) = 0$.

2.2. **How to control these coefficients.** We seek an analytic continuation of $D[\chi]$ in the angle. The reason is elementary. If an entire continuation satisfies

$$|D[\chi](\theta + i\beta)| \le C_\chi e^{2N|\beta|},$$

then shifting the Fourier integral to height β gives

$$|c_\ell[\chi]| \le C_\chi e^{2N|\beta| + \ell\beta}.$$

Letting β tend to the appropriate infinity forces $c_\ell[\chi] = 0$ when $|\ell| > 2N$. The problem is thus to obtain analytic continuation and a growth estimate from our assumptions.

Reflection positivity supplies the continuation. Reflection across $t = 0$ constructs a Hilbert space carrying a Hamiltonian $H \ge 0$ and a commuting momentum P. Comparing with the diagonal reflection gives

$$H \ge |P| \qquad \text{(Section 3)}.$$

For complex translation parameters z_t, z_x, the operator $e^{-z_t H + i z_x P}$ is therefore bounded when $\operatorname{Re} z_t > |\operatorname{Im} z_x|$. Gaussian domination bounds the field insertions between these evolution operators. Together they give analytic moment kernels at separated planar points.

We do not evaluate a Schwartz test at a complex point. Instead, we continue the moment kernel and integrate it against the original tests. Each field costs at most one power of inverse planar separation. An imaginary rotation can enlarge that cost by $e^{|\beta|}$, and rotating the two-form indices costs at most another $e^{|\beta|}$. This accounts for the bound $e^{2N|\beta|}$.

These estimates first apply when the tests have disjoint planar supports. Here and below, the *planar support* is the projection of the support onto the tx-plane. To reach arbitrary tests, we use closedness: a bounded multilinear form that annihilates $d^*\eta$ in each entry is determined by its values on disjoint planar supports. Applying this fact to c_ℓ extends the frequency bound to all tests. Section 4 therefore ends with

$$D[\chi](\theta) = \sum_{\substack{\ell\in 4\mathbb{Z} \\ 0<|\ell|\le 2N}} c_\ell[\chi](e^{i\ell\theta} - 1). \tag{15}$$

The subtraction of 1 incorporates $D[\chi](0) = 0$.

2.3. **How positivity eliminates the remaining coefficients.** At an imaginary angle, the factors $e^{i\ell\theta}$ become real exponentials $e^{-\ell\beta}$. This lets us compare Fourier frequencies by their rates of growth.

The defect itself is not a positive kernel. We must add back the unrotated moment:

$$S_N(\chi_1, \dots, \chi_N) + D[\chi](i\beta).$$

For tests arranged on opposite sides of the reflection plane, the insertion construction expresses this quantity as an inner product. Cauchy–Schwarz then compares a correlation with $a + b$ insertions to two squared norms, with $2a$ and $2b$ insertions. In particular, the largest growing term of the correlation cannot exceed the growth allowed by those norms.

We also have the diagonal reflection. Let ν be the law obtained by rotating the tests by $\pi/4$; at this stage it need not equal μ. The definitions give, directly,

$$D_\nu[\chi](\theta) = D_\mu[\chi](\theta + \pi/4) - D_\mu[\chi](\pi/4). \tag{16}$$

Thus each nonzero Fourier coefficient is multiplied by $e^{i\ell\pi/4}$, which is a sign because $\ell \in 4\mathbb{Z}$.

The coordinate and diagonal reflection directions exchange roles in the tilted frame, so it has the same analytic construction and its own positive inner products. A positive rescaling at $\beta \to +\infty$ isolates the leading coefficient of an even-order moment. That coefficient must be positive in both frames. A change of sign is impossible unless it is zero.

Section 5 combines this sign comparison with Cauchy–Schwarz. A surviving nonzero frequency would force the leading growth at higher moment orders to exceed the bound in (15). Hence every nonzero $c_\ell[\chi]$ vanishes, and so does $D[\chi]$. Gaussian domination then recovers the characteristic function from the moments.

3. $H \geq |P|$

To continue the moments, we need control of spatial momentum by the Hamiltonian. The coordinate reflection constructs these two operators; the diagonal reflection relates them. We prove $H \geq |P|$ by comparing two spectral representations of the same correlation.

Replacing F by $(2C)^{-1/2}F$ reduces the Gaussian bound to $\mathbb{E}_\mu e^{F(\chi)} \leq e^{\|\chi\|_2^2/2}$. We use this normalization until the end. The moment estimate proved in Appendix A is

$$\|F(\chi)\|_{L^p(\mu)} \leq K\sqrt{p}\,\|\chi\|_2 \qquad (p \geq 1),$$

where K is a numerical constant. We allow K to increase between estimates. In particular, the field extends to complex L^2 tests.

3.1. **The reflection Hilbert space.** For a translation vector $a \in \mathbb{R}^d$, put $\tau_a\chi(q) = \chi(q - a)$. On cylinder observables define

$$U(a)F(\chi) = F(\tau_a\chi), \qquad \widehat{T}F(\chi) = F((T^{-1})^*\chi).$$

For each assumed symmetry T, the operator $\widehat{T}$ is unitary on $L^2(\mu)$. Translations are strongly continuous: for real tests,

$$\|e^{iF(\tau_a\chi)} - e^{iF(\chi)}\|_{L^2(\mu)} \leq \|\tau_a\chi - \chi\|_2.$$

The field exponentials have dense linear span. Indeed, a function orthogonal to all of them has finite-dimensional conditional expectations whose associated finite complex measures have zero Fourier transform. Those conditional expectations vanish, and the evaluations generate the Borel sigma-algebra.

Let $\mathcal{A}_+$ be the sigma-algebra generated by fields with tests in $t > 0$, and define the antilinear reflection

$$JA = \overline{\widehat{\Theta}_\circ A}.$$

Reflection positivity gives a positive sesquilinear form

$$\langle [A], [B] \rangle_{\mathcal{H}} = \mathbb{E}_\mu[(JA)B]. \tag{17}$$

For finite sums of field exponentials this is the transpose of the matrix in (5), hence is positive. The estimate

$$|\mathbb{E}_\mu[(JA)B]| \le \|A\|_{L^2(\mu)}\|B\|_{L^2(\mu)}$$

extends the form to $L^2(\mathcal{A}_+,\mu)$. Quotient by its null space and complete to obtain $\mathcal{H}$. The class map is a contraction, the inner product is antilinear in its first entry, and $\Omega = [1]$. An L^2 test supported in $t \ge 0$ also defines a positive-side observable, by approximation with smooth tests in $t > 0$.

3.2. **The translation generators.** For $s \ge 0$, reflection and translation invariance give

$$\|[U(se_0)A]\|_{\mathcal{H}}^2 = \langle [A], [U(2se_0)A]\rangle_{\mathcal{H}} \le \|[A]\|_{\mathcal{H}}\|[U(2se_0)A]\|_{\mathcal{H}}.$$

Iterate and use $\|[U(re_0)A]\|_{\mathcal{H}} \le \|A\|_{L^2(\mu)}$. This proves contractivity and preservation of null vectors. The induced operators are self-adjoint and strongly continuous, so

$$[U(te_0)A] = e^{-tH}[A], \qquad H \ge 0. \tag{18}$$

Translations parallel to $t = 0$ give strongly continuous unitary groups. In particular,

$$[U(xe_1)A] = e^{ixP}[A].$$

They commute with time evolution. Hence H, P have a joint spectral measure E and

$$V(t,x) := e^{-tH+ixP} = \int e^{-t\omega+ixp}\,\mathrm{d}E(\omega,p), \qquad t \ge 0, \quad \omega \ge 0. \tag{19}$$

Every translation fixes Ω. This construction applies in any reflection direction for which invariance and positivity are known; it assumes no continuous rotation symmetry.

3.3. **Comparing the two spectral representations.** We prove the whole-space spectral statement

$$E\{(\omega,p) : \omega < |p|\} = 0. \tag{20}$$

This is the meaning of $H \ge |P|$. The argument compares two spectral representations of the same correlation, one from coordinate reflection and one from diagonal reflection [M, Proposition 18.1].

Fix $L > 0$ and an observable $A \in L^2(\mu)$ measurable in

$$B = \{0 < t < L,\ |x| < L\}.$$

Let ρ be the positive joint spectral measure in $[A]$. For $a, c \ge 0$, put

$$G(a,c) = \mathbb{E}_\mu\big[(JA)U\big(2Le_0 + a(e_0+e_1) + c(e_0-e_1)\big)A\big]\,.$$

The coordinate reconstruction gives, with $z = \omega - ip$,

$$G(a,c) = \int e^{-2L\omega}e^{-az-c\bar z}\,\mathrm{d}\rho(\omega,p). \tag{21}$$

For the diagonal reconstruction, set

$$v_\pm = (e_0 \pm e_1)/\sqrt{2}, \qquad b = L(e_0+e_1)/2,$$

and define

$$A_1 = \widehat{\Theta_\Delta}U(-b)\widehat{\Theta_\circ}A, \qquad A_2 = U(2Le_0 - b)A.$$

Both observables are supported in $t + x > 0$, and both have ordinary L^2 norm $\|A\|_{L^2(\mu)}$. Translating the defining correlation by $-b$ gives the diagonal reflection pairing of A_1 with $U(\sqrt{2}av_+ + \sqrt{2}cv_-)A_2$. Its spectral representation is

$$G(a,c) = \int e^{-a\omega'+icp'}\,\mathrm{d}\rho'(\omega',p'), \qquad \omega' \ge 0, \tag{22}$$

where the factors $\sqrt{2}$ have been absorbed into the spectral variables and

$$\|\rho'\|_{\mathrm{TV}} \le \|A\|_{L^2(\mu)}^2.$$

For a mixed spectral measure, this total-variation bound follows from Cauchy–Schwarz on finite partitions into mutually orthogonal spectral projections. The two Hilbert spaces need not be identified.

Write $Lh(z) = \int_0^\infty h(a)e^{-az}\,\mathrm{d}a$ for $h \in L^1(0,\infty)$. Integrating the two representations against $h(a)\overline{h(c)}$ yields

$$\int e^{-2L\omega}|Lh(\omega - ip)|^2\,\mathrm{d}\rho \le \|A\|^2_{L^2(\mu)} \sup_{r\ge 0}|Lh(r)| \sup_{s\in\mathbb{R}}|Lh(is)|. \tag{23}$$

We therefore seek transforms for which the product of these two suprema stays bounded, while the modulus grows outside the sector $|\arg z| \le \pi/4$.

Complex powers of z magnify its argument: $|z^{-ib}| = e^{b\arg z}$. This suggests the following probes. For $\varepsilon \in \{1,-1\}$ and $j \ge 1$, choose

$$Lh_j(z) = e^{-j}z^{\,1-4i\varepsilon j/\pi}(1+z)^{-3}, \qquad Lh_j(0) = 0, \tag{24}$$

with principal powers in the right half-plane. These are transforms of integrable functions. Indeed, put $\nu = 1 - 4i\varepsilon j/\pi$ and factor

$$z^\nu(1+z)^{-3} = \bigl(1-(1+z)^{-1}\bigr)^\nu (1+z)^{-(3-\nu)}.$$

The first factor is the transform of the finite measure

$$m_\nu = \delta_0 + \sum_{k\ge 1}(-1)^k\binom{\nu}{k}\frac{a^{k-1}e^{-a}}{(k-1)!}\,\mathrm{d}a.$$

Its coefficients are $O_\nu(k^{-2})$, and each displayed density has integral one. The second factor is the transform of

$$q_\nu(a) = \frac{a^{2-\nu}e^{-a}}{\Gamma(3-\nu)} \in L^1(0,\infty).$$

Thus $h_j = e^{-j}m_\nu * q_\nu$ works. Integrability is required separately for each probe, not uniformly in j. The principal logarithms agree in the factorization; continuity gives the transform identity on the imaginary axis and at zero.

On $\operatorname{Re} z \ge 0$, $|z||1+z|^{-3} \le 1$. The two suprema in (23) are therefore at most e^{-j} and e^{j}, respectively. On the sector $\varepsilon\arg z \ge \pi/4 + \delta$, however,

$$|Lh_j(z)|^2 \ge e^{8\delta j/\pi}|z|^2|1+z|^{-6}.$$

It follows that

$$\int_{\varepsilon\arg z\ge\pi/4+\delta} e^{-2L\omega}|z|^2|1+z|^{-6}\,\mathrm{d}\rho \le e^{-8\delta j/\pi}\|A\|^2_{L^2(\mu)}.$$

Letting $j\to\infty$ shows that this sector has zero ρ-measure. Both signs and a sequence $\delta \downarrow 0$ exclude exactly $|p| > \omega$.

Finite sums of positive-time field exponentials with compactly supported tests lie in regions B of this form. Their classes are dense in $\mathcal{H}$. The spectral projection in (20) therefore vanishes on all of $\mathcal{H}$, proving $H \ge |P|$.

4. Analytic continuation of the defect

We now prove the frequency bound for the original defect, with arbitrary tests.

Proposition 4.1. *For every $N \ge 1$ and every tuple of complex L^2 tests,*

$$D[\chi](\theta) = \sum_{\substack{\ell\in 4\mathbb{Z}\\ 0<|\ell|\le 2N}} c_\ell[\chi](e^{i\ell\theta}-1). \tag{25}$$

In particular, $D[\chi]$ extends to an entire function of the angle. Moreover, any nonzero coefficient form c_ℓ, $\ell \ne 0$, has a nonzero value on product tests with compact planar supports strictly ordered in time, after permuting the entries.

The last assertion will let us place a nonzero coefficient across a reflection plane when applying positivity. Its proof uses closedness, just as the passage to arbitrary tests does.

4.1. **Complex translations and separated fields.** Write $z = (z_t, z_x) \in \mathbb{C}^2$ and set

$$q(z) = \operatorname{Re} z_t - |\operatorname{Im} z_x|, \qquad W = \{z : q(z) > 0\}.$$

The spectral inequality of Section 3 gives the extension

$$V(z) = \int e^{-z_t\omega + iz_x p} \, \mathrm{d}E(\omega, p). \tag{26}$$

It is a contraction on $q \geq 0$ and is operator-norm holomorphic on W. Indeed, on $q \geq \delta > 0$, a spectral derivative of total order k is bounded by $\omega^k e^{-\delta\omega} \leq k!\delta^{-k}$, since $|p| \leq \omega$. The semigroup law continues to hold, and $V(z)\Omega = V(z)^*\Omega = \Omega$.

To obtain a moment kernel, first separate the test variables:

$$\chi_j(z, y) = \alpha_j(z) f_j(y), \qquad f_j \in L^2(\mathbb{R}^{d-2}; \Lambda^2\mathbb{C}^d).$$

Here $z = (t, x)$ is real in a test, α_j is a scalar planar test, and f_j retains all two-form components. Finite sums of such products are dense in L^2, also with prescribed planar supports.

For $Z = (z_1, \ldots, z_N)$ and $f = (f_1, \ldots, f_N)$, write $S_N(Z; f)$ for the moment density with the transverse variables smeared by f. Appendix A constructs this density when the successive differences $z_{j+1} - z_j$ lie in W, and proves

$$|S_N(Z; f)| \leq K^N \prod_{j=1}^{N} \|f_j\|_2 \left(\min_{j<N} q(z_{j+1} - z_j) \right)^{-N}. \tag{27}$$

It is holomorphic in the points, complex multilinear in the profiles, and invariant under common translations. For compact planar supports strictly ordered in t,

$$S_N(\alpha_1 f_1, \ldots, \alpha_N f_N) = \int \prod_{j=1}^{N} \alpha_j(z_j) \, S_N(Z; f) \, \mathrm{d}Z. \tag{28}$$

We use $S_0 = 1$ and $S_1 = 0$ as before.

The estimate has a simple source. Gaussian domination makes an insertion in a time slab bounded by $K\|\chi\|_2$. Holomorphic averaging then replaces a small planar support by a point insertion, at the cost of one inverse planar gap. Multiplying N insertions gives (27). The transverse coordinates are never localized, so the exponent is N, independently of d.

4.2. **Continuing the same rotation defect.** For these product tests, changing variables in the rotated moment gives

$$\begin{aligned} D[\chi](\theta) = \int \prod_j \alpha_j(z_j) \, [S_N(R(-\theta)Z; R(\theta)^* f) \\ - S_N(Z; f)] \, \mathrm{d}Z, \end{aligned} \tag{29}$$

initially whenever the supports and their rotations lie in ordered regions. Operations on a tuple are applied entry by entry. The inverse rotation of the points comes from the ordinary pullback on tests. Only the kernel and the finite-dimensional tensor action will be continued in θ; the functions α_j remain at real arguments.

We first join the ordered kernels. Permuting the entries gives every time ordering. Quarter turns give the corresponding orderings with either t or x as time. On a quarter-turned domain, rotate the points to the original domain and apply the inverse pullback to their profiles. These kernels agree wherever their domains overlap. An intersection is convex and contains the real part of each of its points; if nonempty, it contains a real open set. On that set both kernels represent the same moments. Testing against products of smooth functions gives equality there, and the holomorphic identity theorem gives equality on the intersection.

We continue to denote the joined kernel by S_N. Its real domain consists of configurations for which either all t coordinates or all x coordinates are distinct. It is symmetric under simultaneous permutations of points and profiles. Its quarter-turn covariance includes the tensor action:

$$S_N(R(-\pi/2)Z; R(\pi/2)^* f) = S_N(Z; f).$$

This identity uses an assumed symmetry, not the rotation invariance we are proving.

For a complex angle $\theta + i\beta$, there are two contributions to growth. The identities

$$(R(-\zeta)z)_t \pm i(R(-\zeta)z)_x = e^{\pm i\zeta}(z_t \pm iz_x)$$

show that $R(-i\beta)$ preserves W and each quarter-turned ordered domain, with

$$q(R(-i\beta)z) \geq e^{-|\beta|} q(z). \tag{30}$$

For the tensor indices,

$$\|R(\theta + i\beta)^* f\|_2 \leq e^{|\beta|} \|f\|_2. \tag{31}$$

Indeed, $\mathrm{d}t \wedge \mathrm{d}x$ and the forms with two transverse indices are fixed. On each pair of mixed components, the orthogonal complex directions $(\mathrm{d}t \pm i\mathrm{d}x) \wedge \mathrm{d}y_a$ have eigenvalues $e^{\mp i(\theta+i\beta)}$. Their largest modulus is $e^{|\beta|}$.

Take a real configuration whose successive differences make angles less than some $\theta_0 < \pi/4$ with the positive time axis. Its whole complex rotation orbit lies in the joined domains. For each real angle, a quarter-turn frame puts every successive difference within $\pi/4 + \theta_0 < \pi/2$ of that frame's positive time axis; imaginary rotations preserve that domain. Combining (27)–(31) yields

$$\left|S_N(R(-\theta - i\beta)Z; R(\theta + i\beta)^* f)\right| \leq C(Z, f) e^{2N|\beta|}. \tag{32}$$

The same estimate, with a larger constant, holds after subtracting $S_N(Z; f)$.

This is an entire, $\pi/2$-periodic function of the angle. The contour shift in Section 2 therefore removes every frequency with $|\ell| > 2N$. Denote its nonzero coefficient densities by $c_\ell(Z; f)$, $\ell \neq 0$. They will give the already defined $c_\ell[\chi]$ on integration. Since the difference vanishes at angle zero, the expansion can be written as

$$S_N(R(-i\beta)Z; R(i\beta)^* f) = S_N(Z; f) + \sum_{\substack{\ell \in 4\mathbb{Z} \\ 0 < |\ell| \leq 2N}} c_\ell(Z; f)(e^{-\ell\beta} - 1). \tag{33}$$

4.3. **Extending the estimate beyond narrow configurations.** The preceding argument used narrow configurations only to obtain an entire rotation orbit. Its conclusion extends to every ordered domain by holomorphic uniqueness. To see this without assuming a larger domain of continuation, apply

$$\prod_{\substack{\ell \in 4\mathbb{Z} \\ |\ell| \leq 2N}} (\partial_\beta + \ell)$$

to $S_N(R(-i\beta)Z; R(i\beta)^* f)$. For each fixed real β and fixed profiles, the result is holomorphic in Z. It vanishes on the real open set just considered. Every ordered domain contains a reordered, quarter-turned such set, so it vanishes throughout each domain.

The roots of this constant-coefficient equation are distinct. Its solutions are finite sums of the exponentials $e^{-\ell\beta}$, and hence (33) holds on every ordered domain. The coefficients are holomorphic in Z: evaluate the finite sum at as many distinct integer values of β as there are frequencies and invert the resulting Vandermonde matrix. They are bounded multilinear in the profiles locally uniformly in Z, by the same finite inversion and (27). Uniqueness identifies them on overlaps.

Analytic continuation in the angle now gives the local transformation rule

$$c_\ell(R(-\theta)Z; R(\theta)^* f) = e^{i\ell\theta} c_\ell(Z; f), \qquad \ell \neq 0, \tag{34}$$

and the real-angle version of (33), whenever the rotation remains in one of the domains.

These identities also extend the kernel to every configuration of distinct real planar points. The geometric reason is that generic configurations have their whole real rotation orbit in the joined

domain. More precisely, configurations with no two distinct chords perpendicular form a dense, open, rotation-invariant set. An orbit could fail both the t and the x orderings only if it had a horizontal chord and a vertical chord. Thus the local identities continue around the full orbit of every such configuration. Approximation gives the same identities whenever both endpoints lie in the joined real domain.

For an arbitrary distinct configuration Z, choose θ so that $R(-\theta)Z$ has distinct time coordinates; only finitely many angles are forbidden modulo π. Define the coefficient at Z by transporting (34) backwards:

$$c_\ell(Z; f) = e^{-i\ell\theta} c_\ell(R(-\theta)Z; R(\theta)^* f).$$

Then define the moment kernel by the real-angle expansion:

$$S_N(Z; f) = S_N(R(-\theta)Z; R(\theta)^* f) - \sum_{\substack{\ell \in 4\mathbb{Z} \\ 0<|\ell|\le 2N}} c_\ell(Z; f)(e^{i\ell\theta} - 1).$$

The identities already proved make both definitions independent of the choice of θ. One angle works on a neighborhood of Z, so the extensions are real analytic. They agree with the original ordered kernels and retain the finite expansion. We have not yet asserted that this continued kernel represents the original moments on all separated supports; that is the next step.

4.4. **Identifying the continued kernel with the moments.** We prove that (28) holds for arbitrary pairwise disjoint compact planar supports. It suffices to prove it for small discs about each distinct configuration and then use a finite subdivision of the supports.

The identity is known when either all times or all x coordinates are distinct. Induct on the number of pairs of points with equal x coordinate. If there is such a pair, its two times differ. Cut the configuration between these times, at a level containing no point, into an earlier and a later group. Translate the later group horizontally by a small positive distance u_0. Choose u_0 less than every nonzero difference between the original x coordinates, when such differences exist. Cross-group ties disappear and no new ones arise. At least the chosen tie disappears, so the induction hypothesis applies to the translated configuration.

Choose small discs about the original points that preserve the time gap. Translate the later tests horizontally by u. The moment is real analytic in u. In fact, move the separating line to $t = 0$ and leave a gap $2\delta > 0$ between the two groups. If A_-, A_+ are the products of their fields, set

$$\psi_- = [U(-\delta e_0) J A_-], \qquad \psi_+ = [U(-\delta e_0) A_+].$$

These classes exist by the moment bound. Reflection and translation give

$$\mathbb{E}_\mu[A_- U(u e_1) A_+] = \langle \psi_-, V(2\delta, u)\psi_+ \rangle. \tag{35}$$

The right side is holomorphic for $|\operatorname{Im} u| < 2\delta$.

The proposed kernel integral is also real analytic in u. After changing variables, the tests remain on fixed compact supports and only the kernel arguments move. On a compact interval of u, the configurations stay in a compact subset of the distinct-point region. A finite cover by local holomorphic extensions of S_N gives a common positive radius and a uniform bound. Cauchy's estimates justify differentiation under the integral.

The two expressions agree for u near u_0, by induction, and therefore agree at $u = 0$. This proves the local identity and completes the induction. Subdividing any disjoint compact supports into sufficiently small pieces gives (28) on those supports. No rotation invariance has been used in this identification.

We can now integrate the finite expansion. For separated product tests, (29) holds at every real angle, and

$$c_\ell[\chi] = \int \prod_j \alpha_j(z_j) c_\ell(Z; f)\, \mathrm{d}Z \qquad (\ell \neq 0). \tag{36}$$

Thus $D[\chi]$ has the finite expansion (25) on separated supports. Density of product tests gives the same conclusion for arbitrary L^2 tests with those supports.

4.5. **From separated supports to arbitrary tests.** For every ℓ, the definition (14) and the moment bound make c_ℓ a bounded multilinear form on L^2 tests. Closedness implies that it vanishes if one entry is $d^*\eta$: both moments defining D vanish, since pullback commutes with d^*.

We use the following fact, proved in Appendix B.

Lemma 4.2 (Closedness and separated supports)**.** *Let $d \geq 3$, let $O \subset \mathbb{R}^2$ be open, and let B be a bounded N-linear form on $L^2(O \times \mathbb{R}^{d-2}; \Lambda^2\mathbb{C}^d)$. Suppose that B vanishes on pairwise disjoint compact planar supports and whenever one entry is $d^*\eta$, with $\eta \in C_c^\infty(O \times \mathbb{R}^{d-2}; \Lambda^3\mathbb{C}^d)$. Then $B = 0$.*

The disjoint-support condition alone would not suffice: a bilinear form such as $\int \chi(z)\eta(z)\,\mathrm{d}z$ is supported on the diagonal. Closedness rules out this possibility for the two-form defects under consideration.

For $|\ell| > 2N$, the coefficient c_ℓ vanishes on separated supports by the analytic argument. Lemma 4.2 therefore makes it zero on all tests. The same lemma says that a nonzero c_ℓ must be nonzero on some separated supports. By product approximation, one can choose product tests. Their coefficient has the real-analytic density in (36). A nonzero such density is nonzero somewhere with distinct time coordinates, and sufficiently small product neighborhoods there give a nonzero value with strictly time-ordered supports.

For Schwartz tests, Fourier inversion and $D[\chi](0) = 0$ now prove (25). The moment bound extends the identity to all L^2 tests, completing Proposition 4.1. Its finite sum is the desired entire continuation of the original $D[\chi]$. We henceforth use this continuation, not a complex action on arbitrary test functions.

5. Eliminating Fourier coefficients

The defect is now a finite Fourier sum. To show it vanishes, we use reflection positivity on its continued moments, first in the coordinate frame and then in the diagonal frame.

5.1. **Positive inner products at imaginary angles.** The test version of the antilinear reflection is

$$J\chi = \Theta_\circ^*\overline{\chi}, \qquad JF(\chi) = F(J\chi).$$

On tuples it is applied to every entry. Let χ and η be tuples of product tests, each with compact planar supports strictly ordered in the positive half-plane $t > 0$. They may contain different numbers a, b of tests. The construction in Appendix A.7 gives vectors $v_\chi(\beta), v_\eta(\beta) \in \mathcal{H}$ such that

$$\langle v_\chi(\beta), v_\eta(\beta)\rangle = S_{a+b}(J\chi, \eta) + D[J\chi, \eta](i\beta). \tag{37}$$

Here $(J\chi, \eta)$ denotes concatenation of the two tuples. An empty tuple gives Ω. At $\beta = 0$, the vector is the class of the corresponding product of fields.

The formula follows by moving the planar insertion points to $R(-i\beta)z$ and applying $R(i\beta)^*$ to their transverse profiles. These are legitimate operations on the analytic kernels constructed above. Taking adjoints reflects the insertion points and conjugates their profiles; the identities

$$\Theta_\circ\overline{R(-i\beta)z} = R(-i\beta)\Theta_\circ\bar{z}, \qquad \Theta_\circ^*\overline{R(i\beta)^* f} = R(i\beta)^*\Theta_\circ^*\bar{f}$$

then give (37). The appendix verifies the integral and operator identities. No rotation operator on $\mathcal{H}$ is assumed.

Consequently, the right side of (37), as a function of the two tuples, is a positive semidefinite kernel for each real β. In particular,

$$|\langle v_\chi(\beta), v_\eta(\beta)\rangle|^2 \leq \|v_\chi(\beta)\|^2\|v_\eta(\beta)\|^2. \tag{38}$$

The constant moment in (37) is essential: positivity is a statement about the moment, not about its difference from another moment.

5.2. **Changing the reflection plane.** Set $Q = R(\pi/4)$ and define a law ν by

$$\Phi_\nu(\chi) = \Phi_\mu(Q^*\chi).$$

Equivalently, push μ forward under $F \mapsto [\chi \mapsto F(Q^*\chi)]$. We do not assume $\nu = \mu$.

The constructions of Sections 3 and 4 apply to ν. Gaussian domination and closedness are preserved. Translations are conjugated to translations, and quarter turns commute with Q. Reflection positivity and invariance are transported by conjugation; the relevant normals are

$$Q^{-1}e_0 = \frac{e_0 + e_1}{\sqrt{2}}, \qquad Q^{-1}\frac{e_0 + e_1}{\sqrt{2}} = e_1.$$

Thus the coordinate and diagonal directions for ν are available directions for μ. The support conditions are transported as well, since $Q^*\chi$ is supported in the half-space with normal $Q^{-1}v$ when χ is supported in the half-space with normal v. No symmetry mixing the transverse coordinates with the plane was used in the preceding constructions.

The two laws have the same tests and two separate reflection Hilbert spaces. Their defects satisfy

$$D_\nu[\chi](\theta) = D_\mu[\chi](\theta + \pi/4) - D_\mu[\chi](\pi/4), \tag{39}$$

just by expanding the definitions. Hence

$$c^\nu_\ell[\chi] = e^{i\ell\pi/4}c^\mu_\ell[\chi] = (-1)^{\ell/4}c^\mu_\ell[\chi] \qquad (\ell \in 4\mathbb{Z} \setminus \{0\}). \tag{40}$$

This compares the original Fourier coefficients directly. Since the Fourier sums are finite, (39) also holds at complex angles. Each law has the positive inner products (37) in its own Hilbert space.

5.3. **Vanishing of the defect.** For this comparison only, let M_N be the largest $|\ell|$ for which c_ℓ is nonzero at order N, and put $M_N = 0$ when no nonzero frequency survives. The two laws have the same M_N by (40). Proposition 4.1 gives

$$M_0 = M_1 = 0, \qquad M_N \in 4\mathbb{Z}, \qquad 0 \le M_N \le 2N. \tag{41}$$

We compare the leading exponentials in (37).

Suppose $M_{a+b} > 0$. Reality gives a nonzero coefficient at frequency $-M_{a+b}$, possibly after conjugating the tests. By Proposition 4.1, choose a nonzero value on strictly time-ordered compact supports. The coefficients are invariant under simultaneous permutations and common translations of the tests, as follows from the same properties of the moments. Translate a gap between the ath and $(a+1)$st supports to $t = 0$. Reflect and conjugate the earlier group. The resulting tuple has the form $(J\chi, \eta)$ in (37), with a and b positive-time insertions. An empty group is handled by placing all supports on the other side and using Ω.

At this choice of tests,

$$|\langle v_\chi(\beta), v_\eta(\beta)\rangle|^2 \sim |c_{-M_{a+b}}[J\chi, \eta]|^2 e^{2M_{a+b}\beta} \qquad (\beta \to +\infty).$$

The squared norms in (38) are $O(e^{M_{2a}\beta})$ and $O(e^{M_{2b}\beta})$, respectively. Cauchy–Schwarz therefore gives

$$2M_{a+b} \le M_{2a} + M_{2b}. \tag{42}$$

When $M_{a+b} = 0$, the inequality is automatic.

Next fix an even order $2a$ with $M_{2a} > 0$, and restrict both arguments of (37) to tuples of length a. Multiply the kernel by the positive number $e^{-M_{2a}\beta}$ and let $\beta \to +\infty$. Every constant term and every lower exponential disappears. Thus

$$(\chi, \eta) \longmapsto c^\mu_{-M_{2a}}[J\chi, \eta]$$

is positive semidefinite: every finite matrix is a limit of positive semidefinite Gram matrices. The same holds for ν. But the two limiting kernels differ by $(-1)^{M_{2a}/4}$, by (40). If this sign were negative, every finite matrix of the limiting kernel would be both positive and negative semidefinite, hence zero. The preceding placement of a nonzero coefficient, with a insertions on each side, rules this out. Consequently,

$$M_{2a} \in 8\mathbb{Z}. \tag{43}$$

This includes $M_{2a} = 0$. Only the leading coefficient has been isolated as a positive limit; no positivity of the other Fourier coefficients is asserted.

It remains to combine these comparisons. Taking $(a, b) = (j - 1, j + 1)$ in (42) shows that the successive increments of the sequence M_{2j} are nondecreasing. By (43), each increment is a multiple of eight. Since $M_0 = M_2 = 0$, all increments are nonnegative. Any positive increment would make every later increment at least eight, contradicting $M_{2j} \le 4j$ for large j. Thus all even-order maxima vanish. Finally,

$$0 \le 2M_N \le M_0 + M_{2N} = 0.$$

Every nonzero Fourier coefficient of $D[\chi]$ is zero. Equation (25) gives

$$D[\chi](\theta) = 0 \tag{44}$$

for every tuple of L^2 tests and every real angle.

5.4. **Recovering the law.** For a real test χ, Gaussian domination gives $\mathbb{E}_\mu e^{|F(\chi)|} < \infty$. Thus

$$\Phi(\chi) = \sum_{N=0}^{\infty} \frac{i^N}{N!} S_N(\chi, \dots, \chi)$$

converges absolutely. Equation (44) implies $\Phi(R(\theta)^*\chi) = \Phi(\chi)$. Applying this identity to linear combinations of finitely many real tests gives equality of their joint characteristic functions. The evaluations generate the Borel sigma-algebra of $\mathcal{S}_2'$, as recalled in Appendix A; hence the laws agree.

Signed permutations conjugate rotations of the tx-plane to rotations in every coordinate plane. Such rotations generate $SO(d)$: successive plane rotations carry any unit vector to e_{d-1}, and induction applies on its orthogonal complement. The signed permutation group also contains $\Theta_\circ$, of determinant -1. Together with the assumed translations, these symmetries give invariance under all Euclidean isometries.

Finally, let Θ reflect across an arbitrary affine hyperplane, and choose an isometry T carrying $H_\circ$ and its positive side to that hyperplane and its chosen side. Then $\Theta = T\Theta_\circ T^{-1}$. For tests χ_α on that side, put $\eta_\alpha = T^*\chi_\alpha$. They are supported in $t > 0$, and

$$\Phi(\chi_\alpha - \Theta^*\chi_\beta) = \Phi(\eta_\alpha - \Theta_\circ^*\eta_\beta)$$

by invariance and $T^*\Theta^* = \Theta_\circ^* T^*$. Reflection positivity follows from the coordinate reflection form. Restoring the original normalization of F proves Theorem A. □

Appendix A. Field estimates and analytic kernels

We prove the moment and insertion estimates used in Sections 3 and 4, then construct the vectors in (37). The normalization remains $\mathbb{E}_\mu e^{F(\chi)} \le e^{\|\chi\|_2^2/2}$.

A.1. **Moment bounds.** For real χ, apply the Gaussian bound to $u\chi$, $u \in \mathbb{R}$. Expansion at zero and the exponential Markov inequality give

$$\mathbb{E}_\mu F(\chi) = 0, \qquad \mathbb{E}_\mu |F(\chi)|^2 \le \|\chi\|_2^2, \qquad \mu\{|F(\chi)| > r\} \le 2e^{-r^2/(2\|\chi\|_2^2)}.$$

Integrating the tail gives

$$\|F(\chi)\|_{L^p(\mu)} \le K\sqrt{p}\,\|\chi\|_2 \qquad (p \ge 1). \tag{45}$$

Splitting into real and imaginary parts gives the same estimate for complex tests. Hence $\chi \mapsto F(\chi)$ extends continuously from $\mathcal{S}_2$ to complex L^2 two-form tests, with values in every finite $L^p(\mu)$. These extensions agree. Approximation and Fatou's lemma extend the Gaussian bound to real L^2 tests.

Hölder's inequality now gives

$$|S_N(\chi_1, \dots, \chi_N)| \le (K\sqrt{N})^N \prod_{j=1}^{N} \|\chi_j\|_2. \tag{46}$$

Thus the moments used in the proof strategy are well-defined bounded complex multilinear forms.

A.2. **The Borel sigma-algebra.** For completeness, the Borel sigma-algebra of the weak dual $\mathcal{S}_2'$ is generated by evaluations $F \mapsto F(\chi)$. Choose increasing seminorms p_j defining the Schwartz topology and a countable dense set of tests. Every continuous linear functional belongs to some set

$$E_{j,k} = \{F : |F(\chi)| \le k p_j(\chi) \text{ for all } \chi \in \mathcal{S}_2\}, \qquad j,k \ge 1.$$

Each $E_{j,k}$ is measurable for the evaluation sigma-algebra: its inequalities can be checked on the countable dense set. On $E_{j,k}$, equicontinuity makes evaluation at any test a uniform limit of evaluations at that dense set. Its weak topology is therefore induced by countably many evaluations and is second countable. Its Borel sets belong to the restricted evaluation sigma-algebra. Since $\mathcal{S}_2' = \bigcup_{j,k} E_{j,k}$, the same holds on the whole dual. The converse follows from continuity of the evaluations. Equality of characteristic functions thus determines the Borel law.

A.3. **An exponential estimate.** The first step uses the Gaussian bound before differentiating it. This is the source estimate of [M, Lemma 18.3] in the present form conventions.

Lemma A.1 (A source in a time slab)**.** *Let $l > 0$, let h be a complex L^2 two-form supported in $0 \le t \le l$, and let A be a bounded positive-time observable. Then*

$$\|[e^{F(h)} U(l e_0) A]\|_{\mathcal{H}} \le e^{\|\operatorname{Re} h\|_2^2/2} \|[A]\|_{\mathcal{H}}. \tag{47}$$

Proof. Write $U_l = U(l e_0)$. Reflection and a common translation give

$$\begin{aligned} \|[e^{F(h)} U_l A]\|_{\mathcal{H}}^2 &= \langle [A], [e^{F(h^\#)} U_{2l} A] \rangle_{\mathcal{H}}, \\ h^\# &= \tau_{l e_0}(\Theta_\circ^* \bar{h} + h). \end{aligned} \tag{48}$$

The two parts of $h^\#$ occupy adjacent slabs, with disjoint interiors. Hence

$$\|\operatorname{Re} h^\#\|_2^2 = 2\|\operatorname{Re} h\|_2^2, \qquad \tau_{2l e_0}(\Theta_\circ^* \overline{h^\#}) = h^\#.$$

First suppose that a source k of width L satisfies $\tau_{L e_0}(\Theta_\circ^* \bar{k}) = k$. Its m adjacent copies,

$$k^{(m)} = \sum_{j=0}^{m-1} \tau_{jL e_0} k,$$

have the same symmetry at width mL, and $(k^{(m)})^\# = k^{(2m)}$. Set

$$b = \|[A]\|_{\mathcal{H}}, \qquad a_r = \|[e^{F(k^{(2^r)})} U_{2^r L} A]\|_{\mathcal{H}}.$$

Equation (48) gives $a_r^2 \le b a_{r+1}$. Therefore

$$a_0 \le b^{1-2^{-r}} a_r^{2^{-r}}.$$

Gaussian domination of the reflected source and its adjacent copies gives the ordinary bound

$$a_r^2 \le \|A\|_\infty^2 \exp(2^r \|\operatorname{Re} k\|_2^2).$$

Let $r \to \infty$. We obtain $a_0 \le b e^{\|\operatorname{Re} k\|_2^2/2}$. If $b = 0$, the first Cauchy–Schwarz inequality already gives $a_0 = 0$.

Apply this conclusion to $k = h^\#$, whose width is $2l$, and use (48) once more. Since $\|\operatorname{Re} h^\#\|_2^2 = 2\|\operatorname{Re} h\|_2^2$, the result is (47). □

A.4. **A bounded insertion operator.** For a test χ supported in $|t| \le \delta$, define, initially on bounded positive-time observables,

$$B_\delta(\chi)[A] = [F(\tau_{\delta e_0}\chi) U(2\delta e_0) A]. \tag{49}$$

The source has been shifted into $0 \le t \le 2\delta$. Apply Cauchy's estimate to the entire Hilbert-valued map

$$z \longmapsto [e^{zF(\tau_{\delta e_0}\chi)} U(2\delta e_0) A].$$

Its power series converges locally in ordinary $L^2(\mu)$ by Gaussian domination, before passing to the quotient. Lemma A.1, applied on the circle $|z| = \|\chi\|_2^{-1}$, gives

$$\|B_\delta(\chi)\| \le K\|\chi\|_2. \tag{50}$$

For real χ, one may take $K = \sqrt{e}$; splitting a complex test into real and imaginary parts gives a numerical constant for complex χ. The estimate kills null vectors, so $B_\delta(\chi)$ extends to a bounded operator on $\mathcal{H}$. Truncation in L^4 extends (49) to positive-time observables lying in every finite L^p.

The centered notation makes the reflection and translation identities simple. With $U_\delta = U(\delta e_0)$,

$$\langle [A'], B_\delta(\chi)[A]\rangle_{\mathcal{H}} = \mathbb{E}_\mu[(JU_\delta A')F(\chi)U_\delta A], \tag{51}$$

$$B_\delta(\chi)^* = B_\delta(\Theta_\circ^*\bar{\chi}), \tag{52}$$

$$V(r+t,x)B_\delta(\chi)V(r-t,-x) = B_{\delta+r}(\tau_{(t,x,0)}\chi), \qquad r \ge |t|. \tag{53}$$

The zero in the last formula denotes the transverse translation. These identities follow from (49) by reflection and translation. For example, both sides of (53) act on $[A]$ as

$$[F(\tau_{(\delta+r+t,x,0)}\chi)U(2(\delta+r)e_0)A].$$

Thus they are identities of bounded operators, not formal identities involving point fields.

If χ is supported in $|t-c| \le \delta_0$ and $|c| + \delta_0 \le \delta$, the same identity gives the localization formula

$$B_\delta(\chi) = e^{-(\delta+c-\delta_0)H}B_{\delta_0}(\tau_{-ce_0}\chi)e^{-(\delta-c-\delta_0)H}. \tag{54}$$

It separates the time used by the field from the unused time on either side.

We have converted Gaussian domination into the uniform operator bound (50). The next step is to shrink a field's planar support while keeping positive separation from the other insertions.

A.5. **Localizing an insertion in the plane.** Fix a transverse two-form profile $f \in L^2(\mathbb{R}^{d-2}; \Lambda^2\mathbb{C}^d)$. Let $e = (1,0)$ in the (t,x)-plane and put

$$j_\epsilon(z) = (\pi\epsilon^2)^{-1}\mathbf{1}_{\{|z|<\epsilon\}}.$$

For $q(\xi), q(\eta) > \epsilon$, define

$$A_{f,\epsilon}(\xi,\eta) = V(\xi - \epsilon e)B_\epsilon(j_\epsilon f)V(\eta - \epsilon e).$$

The slab estimate alone would grow as ϵ^{-1}. Analyticity lets us replace the shrinking support scale by the fixed separation from the neighboring insertions.

Set $q_0 = \min(q(\xi), q(\eta))$. We claim

$$\|A_{f,\epsilon}(\xi,\eta)\| \le \frac{K\|f\|_2}{q_0 - \epsilon}. \tag{55}$$

To prove this, put $r = (q_0 - \epsilon)/2$ and, for unit vectors $u, v \in \mathcal{H}$, consider

$$h_\epsilon(\zeta) = \langle u, A_{f,\epsilon}(\xi + \zeta, \eta - \zeta)v\rangle.$$

It is holomorphic where $|\operatorname{Re}\zeta_t| < r$ and $|\operatorname{Im}\zeta_x| < r$. Write $\zeta = z + ib$ with $z, b \in \mathbb{R}^2$. For $|b_x| < r$, the outer factors

$$V(\xi - (r+\epsilon)e + ib), \qquad V(\eta - (r+\epsilon)e - ib)$$

are contractions. Between them, (53) identifies the remaining operator as $B_{r+\epsilon}((\tau_z j_\epsilon)f)$. The outer vectors do not depend on z. Consequently, for $\alpha \in L^2((-r,r)^2)$,

$$\left|\int \alpha(z)h_\epsilon(z+ib)\,\mathrm{d}z\right| \le K\|f\|_2\|\alpha * j_\epsilon\|_2 \le K\|f\|_2\|\alpha\|_2.$$

Duality bounds the real-square integral of $|h_\epsilon(z+ib)|^2$ by $K^2\|f\|_2^2$. Apply the mean-value inequality on a complex polydisc of radius $r' < r$ and integrate over the imaginary square. This gives

$$|h_\epsilon(0)|^2 \le (\pi r'^2)^{-2}(2r)^2K^2\|f\|_2^2.$$

Let $r' \uparrow r$ and take the supremum over u, v. This proves (55).

We next identify the limit as $\epsilon \downarrow 0$. Fix $\delta = q_0/2$, take $0 < \epsilon < \delta/2$, and let α be supported in $(-\delta/2, \delta/2)^2$. The covariance identity gives

$$\begin{aligned} &\int \alpha(z) A_{f,\epsilon}(\xi + z, \eta - z)\, \mathrm{d}z \\ &\qquad = V(\xi - \delta e) B_\delta((\alpha * j_\epsilon) f) V(\eta - \delta e). \end{aligned} \tag{56}$$

The slab width and the outer factors are independent of ϵ. By (50), the right side converges in operator norm as $\alpha * j_\epsilon \to \alpha$ in L^2.

For fixed u, v, the functions h_ϵ are locally uniformly bounded by (55). Montel's theorem gives normally convergent subsequences. Equation (56) identifies their integrals against every real-square test, so any two limits agree on that square. The holomorphic identity theorem makes them agree throughout the polydisc. Thus every sequence $\epsilon \downarrow 0$ has the same scalar limit at its center. The limits are bounded sesquilinear forms in u, v, and hence define an operator $A_f(\xi, \eta)$ with

$$\|A_f(\xi, \eta)\| \le \frac{K \|f\|_2}{\min(q(\xi), q(\eta))}. \tag{57}$$

This is a point insertion only in the plane; the transverse variables are still smeared by f.

Write $\xi^\dagger = (\bar{\xi}_t, -\bar{\xi}_x)$. Spectral calculus gives $V(\xi)^* = V(\xi^\dagger)$. The translation and adjoint identities pass to the weak limit:

$$V(\kappa) A_f(\xi, \eta) V(\kappa') = A_f(\kappa + \xi, \eta + \kappa'), \tag{58}$$

$$A_f(\xi, \eta)^* = A_{\Theta^*_\circ \bar{f}}(\eta^\dagger, \xi^\dagger). \tag{59}$$

The shift identity also proves norm holomorphy. Near (ξ_0, η_0), choose a small $a > 0$ and factor the operator as

$$V(\xi - \xi_0 + ae) A_f(\xi_0 - ae, \eta_0 - ae) V(\eta - \eta_0 + ae).$$

Both outer arguments lie in W on a neighborhood of that point.

For a compactly supported $\alpha \in L^2(\mathbb{R}^2)$ with $|t| < \delta$ on its support,

$$B_\delta(\alpha f) = \int \alpha(z) A_f(\delta e + z, \delta e - z)\, \mathrm{d}z. \tag{60}$$

The integral converges in operator norm by (57). It follows by integrating the regularized identity: the smeared side converges because $\alpha * j_\epsilon \to \alpha$ in L^2, and the matrix elements on the other side converge under a uniform bound on the compact support.

We will also use the fact that an inner time gap can be split in any way. If $a, b, a', b' \in W$ and $a + b = a' + b'$, then

$$A_f(\xi, a) A_{f'}(b, \eta) = A_f(\xi, a') A_{f'}(b', \eta). \tag{61}$$

Choose a small $\delta > 0$ such that all four inner arguments remain in W after subtracting δe. The left side becomes

$$A_f(\xi, \delta e) V(a + b - 2\delta e) A_{f'}(\delta e, \eta),$$

which depends only on $a + b$.

A.6. **Constructing the ordered moment kernels.** For $N \ge 2$ and a configuration $Z = (z_1, \dots, z_N) \in (\mathbb{C}^2)^N$ with $\Delta_j = z_{j+1} - z_j \in W$, define

$$\begin{aligned} S_N(Z; f) = \Big\langle \Omega, A_{f_1}(\kappa, \Delta_1/2) \Big[\prod_{j=2}^{N-1} A_{f_j}(\Delta_{j-1}/2, \Delta_j/2) \Big] \\ A_{f_N}(\Delta_{N-1}/2, \kappa) \Omega \Big\rangle. \end{aligned} \tag{62}$$

The product is ordered by increasing j and is the identity for $N = 2$. The choice of $\kappa \in W$ does not matter: the shift identity and $V\Omega = V^*\Omega = \Omega$ allow any two choices to be enlarged to a common positive-time value. Put $S_0 = 1$ and $S_1 = 0$, the latter agreeing with the mean-zero one-point function.

These kernels are holomorphic, invariant under common translations, and complex multilinear in the transverse profiles. Choose κ with sufficiently large positive time component and apply (57) to each factor. This proves (27).

To verify (28), let $\chi_j = \alpha_j f_j$ and enclose their supports in separated slabs $(c_j - \delta_j, c_j + \delta_j)$. After a common translation, all slabs lie in positive time. Repeatedly applying (49) and (54) expresses the class of their product as

$$e^{-(c_1-\delta_1)H} B_{\delta_1}(\tau_{-c_1 e_0}\chi_1) \left[\prod_{j=2}^{N} e^{-(c_j-\delta_j-c_{j-1}-\delta_{j-1})H} B_{\delta_j}(\tau_{-c_j e_0}\chi_j)\right] \Omega.$$

Insert (60) in each factor. The gaps are positive, so the operator integrals are uniformly bounded on the compact supports. Adjacent semigroups combine into the difference $z_{j+1} - z_j$, and (61) splits each difference equally. Pairing with Ω gives (28).

A.7. **Products of fields at imaginary angles.** We now verify (37). For ordered complex points $Z = (z_1, \dots, z_a)$ with $z_1 \in W$ and $\Delta_j = z_{j+1} - z_j \in W$, define

$$\Psi(Z; f) = A_{f_1}(z_1, \Delta_1/2) \cdots A_{f_a}(\Delta_{a-1}/2, \kappa)\Omega.$$

For $a = 1$, use $A_{f_1}(z_1, \kappa)\Omega$; the empty tuple gives Ω. The choice of $\kappa \in W$ is irrelevant, as for the moment kernel. These vectors are holomorphic in their points and multilinear in their profiles. On strictly ordered positive-time supports, the preceding localization argument gives

$$\left[\prod_{j=1}^{a} F(\alpha_j f_j)\right] = \int \prod_j \alpha_j(z_j)\Psi(Z; f)\,\mathrm{d}Z.$$

All integrals here and below are Hilbert-space integrals; the insertion estimates bound the integrands uniformly on the compact supports.

Taking adjoints, reversing the operator order, and using (61) gives

$$\begin{aligned}\langle \Psi(Z; f), \Psi(Z'; g)\rangle = S_{a+b}(&\Theta_\circ \bar{z}_a, \dots, \Theta_\circ \bar{z}_1, z'_1, \dots, z'_b;\\ &\Theta_\circ^* \bar{f}_a, \dots, \Theta_\circ^* \bar{f}_1, g_1, \dots, g_b).\end{aligned} \tag{63}$$

The first group is reversed so that all displayed points are in the required order. Subsequent permutation symmetry allows this reversal to be suppressed when smearing.

For a tuple $\chi_j = \alpha_j f_j$ on ordered compact positive-time supports, define

$$v_\chi(\beta) = \int \prod_j \alpha_j(z_j)\Psi(R(-i\beta)Z; R(i\beta)^* f)\,\mathrm{d}Z. \tag{64}$$

This is well defined for every real β: imaginary rotations preserve W, the first time gap, and the successive time gaps. On compact β intervals the estimates give a common bound. At $\beta = 0$, it is the class of the product of fields.

Apply (63) under the two integrals defining $\langle v_\chi(\beta), v_\eta(\beta)\rangle$. The reflection identities displayed in Section 5 turn its first group of arguments into the imaginary rotation of the reflected original group, with profiles $R(i\beta)^*\Theta_\circ^* \bar{f}$. Thus the resulting integral is the analytic continuation of

$$S_{a+b}(R(\theta)^* J\chi, R(\theta)^*\eta) = S_{a+b}(J\chi, \eta) + D[J\chi, \eta](\theta)$$

to $\theta = i\beta$.

More explicitly, for real θ near zero the ordered density identity proves this equality of integrals. Both sides are holomorphic in a connected vertical strip about the imaginary axis: choose its real width small enough to preserve the positive gaps, and then use preservation of W by imaginary rotations. The compact-support estimates justify integration there. Holomorphic uniqueness and the entire continuation of D give (37) at every real β. Empty tuples follow from the vacuum identities.

We have obtained both ingredients used in the main argument: the kernel estimate (27) and positive inner products for the continued moments.

Appendix B. Closedness excludes contact terms

We prove that the closedness tests prevent a bounded multilinear defect from being concentrated on coincident planar points. Applied to each Fourier coefficient, this gives both the passage to arbitrary tests and the detection of every nonzero coefficient on separated supports.

Proof of Lemma 4.2. We induct on N, for all open sets O simultaneously. Compact planar truncations are dense. At $N = 1$, the first hypothesis already gives the conclusion.

Bilinear forms. Represent a bounded bilinear form as

$$B(\chi, g) = \int_{O\times\mathbb{R}^{d-2}} \sum_{i<j} \chi_{ij}(Tg)_{ij}$$

with a bounded complex-linear operator T. Vanishing on disjoint compact planar supports gives $\mathbf{1}_K T \mathbf{1}_{K'} = 0$ for disjoint compact K, K'. Inner approximation and L^2 convergence extend this to disjoint Borel sets. Applying it to a set and its complement shows that T commutes with every bounded planar multiplier h. Therefore

$$B(h\, d^*\eta, g) = B(d^*\eta, hg) = 0. \tag{65}$$

We will show that these null tests span every two-form component locally.

Fix a compact $K \subset O$, choose $\rho \in C_c^\infty(O)$ equal to one near K, and take $\psi \in C_c^\infty(\mathbb{R}^{d-2})$. For a transverse index $2 \le a \le d-1$, the codifferentials of

$$\rho\psi\, \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y_a, \qquad t\rho\psi\, \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y_a, \qquad x\rho\psi\, \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y_a$$

restrict on $K \times \mathbb{R}^{d-2}$ to

$$\begin{aligned} &-(\partial_a\psi)\, \mathrm{d}t \wedge \mathrm{d}x, \\ &-t(\partial_a\psi)\, \mathrm{d}t \wedge \mathrm{d}x - \psi\, \mathrm{d}x \wedge \mathrm{d}y_a, \\ &-x(\partial_a\psi)\, \mathrm{d}t \wedge \mathrm{d}x + \psi\, \mathrm{d}t \wedge \mathrm{d}y_a. \end{aligned}$$

For a fixed a, the profiles $\partial_a\psi$ are dense in $L^2(\mathbb{R}^{d-2})$. An L^2 function orthogonal to all of them has distributional derivative zero in y_a. It is therefore constant in that variable for almost every choice of the others, and square integrability on the infinite line makes it zero.

Equation (65), with bounded planar h supported in K, therefore annihilates the whole $\mathrm{d}t \wedge \mathrm{d}x$ component. Subtracting this component from the next two expressions annihilates $\mathrm{d}x \wedge \mathrm{d}y_a$ and $\mathrm{d}t \wedge \mathrm{d}y_a$ for every a.

For $d \ge 4$, take $2 \le a < b \le d-1$. The codifferential of

$$t\rho(z)\psi(y)\, \mathrm{d}t \wedge \mathrm{d}y_a \wedge \mathrm{d}y_b$$

restricts on $K \times \mathbb{R}^{d-2}$ to

$$-\psi\, \mathrm{d}y_a \wedge \mathrm{d}y_b + t(\partial_a\psi)\, \mathrm{d}t \wedge \mathrm{d}y_b - t(\partial_b\psi)\, \mathrm{d}t \wedge \mathrm{d}y_a.$$

The last two components have just been annihilated, so the first is annihilated as well. These are all the components. Density of product tests and compact planar truncation prove $B = 0$ in the bilinear case.

Partial diagonals. Let $N \ge 3$. Fix one entry over a compact planar set K and apply induction to the other $N-1$ entries on $O \setminus K$. The disjoint-support hypothesis is inherited because those supports avoid K. The null-test hypothesis is inherited by extending compactly supported three-forms by zero. Hence B vanishes whenever one entry is over K and every other entry is outside K.

Next fix any nonempty group of entries over K and regard B as a lower-order form in the remaining group on $O \setminus K$. If the latter entries have pairwise disjoint compact supports, any one of them is disjoint from every other entry, including the fixed group. The one-entry conclusion therefore applies. Induction gives vanishing whenever the entries split into two nonempty groups with disjoint compact planar supports. Overlaps within a group are allowed.

The full diagonal. It remains to consider entries with a common compact planar support. Partition the plane into half-open squares Q of side ϵ and expand each entry in its restrictions $\mathbf{1}_Q\chi_i$. Every term involving different squares vanishes by the two-group conclusion, first approximating inside the squares by disjoint compact subsets and then passing to the L^2 limit. Thus

$$\begin{aligned}|B(\chi_1,\dots,\chi_N)| &\le \|B\| \sum_Q \prod_i \|\mathbf{1}_Q\chi_i\|_2 \\ &\le \|B\|\|\chi_1\|_2\|\chi_2\|_2 \sup_Q \|\mathbf{1}_Q\chi_3\|_2 \prod_{i=4}^{N} \|\chi_i\|_2 .\end{aligned}$$

The supremum tends to zero: the function $z \mapsto \int_{\mathbb{R}^{d-2}} |\chi_3(z,y)|^2 \,\mathrm{d}y$ is integrable, and the square areas tend to zero. Density completes the induction. □

The dimension restriction enters in the first bilinear step, where a transverse derivative supplies the $\mathrm{d}t \wedge \mathrm{d}x$ component. In dimension two there are no test three-forms, and $B(\chi, g) = \int \chi_{01} g_{01}$ would satisfy the two hypotheses without vanishing. Thus this no-contact lemma does require $d \ge 3$.

References

[BEG] J. Bros, H. Epstein, and V. Glaser, *On the connection between analyticity and Lorentz covariance of Wightman functions*, Commun. Math. Phys. **6** (1967), 77–100. DOI: 10.1007/BF01654126.

[S] R. F. Streater, *Connection between the spectrum condition and the Lorentz invariance of* $P(\phi)_2$, Commun. Math. Phys. **26** (1972), 109–120. DOI: 10.1007/BF01645698.

[O] E. P. Osipov, *Connection between the spectrum condition and the Lorentz invariance of the Yukawa*$_2$ *quantum field theory*, Commun. Math. Phys. **57** (1977), 111–116. DOI: 10.1007/BF01625770.

[So] A. D. Sokal, *An alternate constructive approach to the* φ^4_3 *quantum field theory, and a possible destructive approach to* φ^4_4, Ann. Inst. H. Poincaré A **37** (1982), 317–398.

[M] *OS axioms for the field strength tensor in three-dimensional electrodynamics*, supplied manuscript `qed3_manuscript(2).pdf`, §18, pp. 104–123.